

FLIGHT TEST # 1

DATE: 11/22/2020

OBJECTIVES

- A successful first flight of the OpenUAS aircraft, including:
 - Sustained flight
 - Verify all aircraft controls function nominally in flight
 - Verify all sensors are functioning during flight
 - Collect flight performance data (climb rate, cruise velocity, descent rate, stall velocity)
- Test flight in PX4 modes such as Position, Stabilized, Hold, Return and Mission modes

MANUAL SUCCESS CRITERIA

- Flight proceeds according to the test procedure
- Flight occurs without any damage to aircraft
- All control surfaces respond as expected
- All flight data correctly displayed and saved in QGroundControl
- Electronics behave as expected and provide reasonable data

STABILIZED SUCCESS CRITERIA

- Aircraft reaches and maintains steady, level flight successfully when sticks are centered

POSITION SUCCESS CRITERIA

- Aircraft maintains cruise throttle and heading when all sticks are centered
- Aircraft completes mission within the estimated time of completion given in QGroundControl

HOLD SUCCESS CRITERIA

- Aircraft loiters over the location where hold mode was turned on
- Aircraft completes mission within the estimated time of completion given in QGroundControl

MISSION SUCCESS CRITERIA

- A successful mission implies the apprentice autonomously executes the following mission successfully:
 - Aircraft maintains cruising speed set in QGroundControl
 - Aircraft reaches successfully changes to the set altitude between waypoints
 - Aircraft climbs to 50 feet as it reaches waypoint two
 - Aircraft holds for 10 seconds at waypoint 2
 - Aircraft climbs to 100 feet as it reaches waypoint 3
 - Aircraft loiters for 30 seconds at waypoints 3
 - Aircraft maintains altitude between waypoints 3 & 4
 - Upon reaching waypoint 4, Aircraft immediately approaches planned home waypoint
 - Upon reaching planned home waypoint, aircraft loiters
- Craft completes mission within the estimated time of completion given in QGroundControl

SUPPLIES

- | | |
|---|--|
| ☐ OpenUAS | ☐ Receiver antenna for computer |
| ☐ Taranis (fully charged) | ☐ Tape |
| ☐ Battery (fully charged, plus spare) | ☐ Zip-ties |
| ☐ Launch Rail | ☐ Anemometer |
| ☐ Computer to run QGroundControl(fully charged, one spare) | ☐ Tool kit |
| ☐ Cell phone with cellular data (fully charged, one spare) | ☐ Spare propellers |

LOCATION:

Central Iowa Aeromodelers Flying Field

TEST SYSTEM:

OpenUAS

TEST CONDITIONS:

Wind W @ 7-10 mph, Temp ~ 30 deg
Unlimited visibility & clear skies

PROCEDURES

1. Complete Preflight Checklist

- ☐ [Electronics Preflight Checks & Arming Procedures](#) completed
- ☐ UAS is ready to fly
- ☐ Team is briefed on safety concerns
- ☐ Team is briefed on roles and objectives
- ☐ Final Weather check

2. Have OpenUAS positioned on launch rail pointed into the wind and “armed”

- ☐ OR, hand launcher holds aircraft pointed into the wind

3. Pilot input full throttle, aircraft is released from the launch rail

- ☐ OR, Pilot input full throttle and hand launcher runs with aircraft and tosses straight ahead.

4. Climb at 100% power to a safe altitude

5. Decrease throttle to ~75%

6. Adjust trims on Taranis as needed

7. Perform a left/right turn and establish a rectangular pattern

8. Test aircraft response for elevator, rudder, & aileron inputs at ½ and full deflection

- ☐ Visually verify the correct response for each maneuver

9. Slow aircraft to just above minimum controllable speed and re-test control inputs

- ☐ Visually verify the correct response for each maneuver

10. Perform a stall at a low throttle setting and again with the throttle fully closed

11. Perform all previous tests again with Flaperons at position 1 & at position 2

12. Perform an approach to landing followed by a low pass over the runway and go-around to assess landing characteristics

13. If flight characteristics allow, start autonomous tests:

Stabilized:

14. Set Cruise throttle in QGroundControl
15. Ensure the home location is set in QGroundControl
16. Set SB switch to lowest position on the Taranis
17. Set flight mode to Stabilized in QgroundControl
18. Verify Stabilized Success Criteria
19. Switch Pixhawk to manual mode by flipping switch SB to its highest position

20. Verify manual control is regained and proceed to other tests or land

Position:

21. Set SB switch to lowest position on the Taranis
22. Set flight to position mode in QgroundControl
23. Verify Position success Criteria Visually and through QGroundControl
 - ☐ Check that the roll, pitch, and yaw headings do not change
 - ☐ Check that throttle is at the set cruise throttle
24. Switch pixhawk back to manual mode
25. Verify manual control is regained and proceed to other tests or land

Hold:

26. Set SB switch to lowest position on the Taranis
27. Set flight mode to hold in QgroundControl
28. Verify Hold success Criteria visually and through QGroundControl
 - ☐ Verify that UAS loiters at the location hold mode was turned on visually and through the GPS Screen in Qgroundcontrol
29. Switch pixhawk back to manual mode
30. Verify manual control is regained and proceed to other tests or land

Return:

31. Set Home location in QGroundControl
32. Set SB switch to lowest position on the Taranis
33. Set mission mode to Return in QGroundControl
34. Verify the Pixhawk flies to the home location and loiters there visually and through the GPS screen
35. Switch Pixhawk to Manually mode
36. Verify manual control is regained and proceed to other tests or land

Mission:

37. Set Mission in QGroundControl
38. Manually take off
39. Fly to waypoint one GPS location
 - ☐ Verify location via map view on QGroundControl
40. Flip the SB switch into the lowest position to switch the pixhawk into mission mode
41. Verify Mission success criteria
42. Once UAS has reached planned home, switch SB into Highest Location to switch the Pixhawk back into manual flight mode
43. Verify manual control is regained

44. Set up and approach and land UAS on the grass

45. After the last flight disconnect in the following order:
 - ☐ Disarm Pixhawk via QgroundControl
 - ☐ Flip ESC power switch to OFF position
 - ☐ Disconnect the battery
 - ☐ Turn off Taranis
 - ☐ Measure battery voltage
 - ☐ Assess aircraft for damage

TEST DATA

- Flight logs in Google Drive: Flight Test/2020-11-22 Open UAS flight/Flight Logs [here](#)
- Media in Google Drive: Flight Test/2020-11-22 Open UAS flight/media [here](#) and Youtube: [here](#)

DEBRIEF

- Total Flight Time: 2 min 40 sec
- Battery level pre-test: 12 V | - Battery level post-test: 11.4 V | - Power Consumption: _____
- Objectives met:
 - Achieved sustained flight of OpenUAS
 - All control surfaces functioned
 - Performed Stabilized and Position flight modes
- Test items that went well:
 - Hand launch worked well
 - OpenUAS had ample performance with large motor/prop/battery
 - Stabilize mode greatly increased stability/flyability of OpenUAS
- Test Items that did not go well/need to be improved:
 - Launch rail did not work, carrier was too flimsy
 - OpenUAS was less stable than anticipated in all axis
 - Ailerons more sensitive and have more deflection than necessary
 - Wind gust caused a loss of airspeed in a turn resulting in a spin. Spin was unrecoverable within the altitude the OpenUAS entered it from, resulting in a crash landing from an induced flat spin.
 - Current Battery monitoring methods are not reliable
 - Measured stall velocities didn't seem to line up with theoretical stall velocity from the CFD